

Car Price Prediction System (MSRP Estimation)

Dataset – [CAR_MSRP](#)

Problem Statement

The automobile market includes a wide variety of cars with different specifications, brands, and features, making price estimation complex. Accurately predicting the Manufacturer's Suggested Retail Price (MSRP) is essential for both buyers and sellers.

The challenge is:

Can we predict the price of a car based on its features such as engine specifications, fuel type, brand, and other attributes?

Objective

Build a **regression model** that predicts the **MSRP (price)** of a car using its technical and categorical features.

Business Impact

- Helps customers make informed purchasing decisions
 - Assists dealerships in pricing strategies
 - Enables competitive market analysis
 - Supports automotive companies in product positioning
 - Improves recommendation systems for car platforms
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Target Variable

- Column Name: **MSRP**

- **Description:** Manufacturer's Suggested Retail Price of the car
- **Type:** Continuous (Numerical)

👉 This is a **regression problem**



Dataset Overview

- **Records:** Thousands of car entries
 - **Features:** 10–15+ input features
 - **Type:** Numerical + Categorical
 - **Domain:** Automotive / Pricing Analytics
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Column Information (Detailed Explanation)

1. **Make**
 - Type: Categorical
 - Description: Brand of the car (e.g., Toyota, BMW, Ford)
2. **Model**
 - Type: Categorical
 - Description: Specific model name of the car
3. **Year**
 - Type: Numerical
 - Description: Manufacturing year of the car
4. **Engine Fuel Type**
 - Type: Categorical
 - Description: Type of fuel used (Petrol, Diesel, Electric, Hybrid)
5. **Engine HP**
 - Type: Numerical
 - Description: Horsepower of the engine
6. **Engine Cylinders**
 - Type: Numerical
 - Description: Number of cylinders in the engine
7. **Transmission Type**
 - Type: Categorical
 - Description: Type of transmission (Automatic, Manual)
8. **Driven Wheels**
 - Type: Categorical
 - Description: Drivetrain type (FWD, RWD, AWD)
9. **Number of Doors**

- Type: Numerical
 - Description: Number of doors in the car
 - 10. **Vehicle Size**
 - Type: Categorical
 - Description: Compact, Midsize, Large
 - 11. **Vehicle Style**
 - Type: Categorical
 - Description: Sedan, SUV, Coupe, Hatchback, etc.
 - 12. **highway MPG**
 - Type: Numerical
 - Description: Fuel efficiency on highways
 - 13. **city MPG**
 - Type: Numerical
 - Description: Fuel efficiency in city
 - 14. **Popularity**
 - Type: Numerical
 - Description: Brand popularity score
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Target Column

- 15. **MSRP**
 - Type: Numerical (Continuous)
 - Description: Price of the car
 - Role: Target Variable